Transformations

NC.8.G.1 • NC.8.G.2 • NC.8.G.3

1 Rigid Transformations: Same Size, Same Shape

Translation	**slide**	every point moves the same way
Reflection	**flip**	mirror image over a line
Rotation	**turn**	spin around a point (the origin)

Image = the new figure. Use **prime**: $A \to A'$ ("A prime").

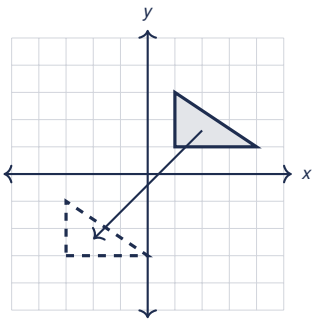
Rigid motions keep the same:

✓ side lengths ✓ angle measures

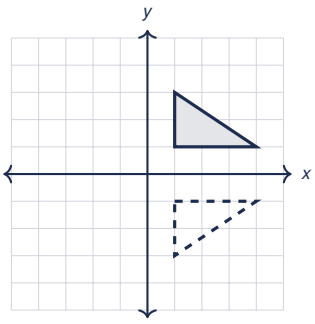
✓ parallel lines stay parallel

Rigid motions only → the figures are **CONGRUENT** ($\cong$).

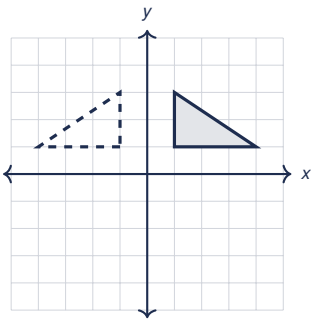
2 Coordinate Rules (shaded = original, dashed = image)



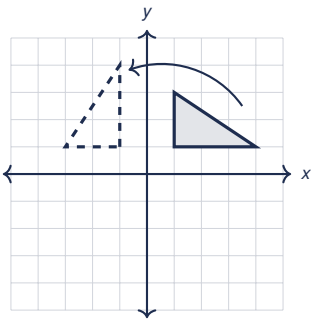
Translate left 4, down 4



Reflect over x-axis



Reflect over y-axis

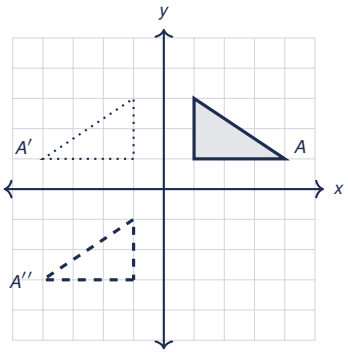


Rotate 90° counterclockwise

Transformation	Rule	Example: $P(2, 3)$	Memory tip
Translation	$(x, y) \to (x + a,\ y + b)$	right 2, down 4: $(4, -1)$	right/up: add left/down: subtract
Reflect over x-axis	$(x, y) \to (x,\ -y)$	$(2, -3)$	y changes sign
Reflect over y-axis	$(x, y) \to (-x,\ y)$	$(-2, 3)$	x changes sign
Rotate 90° counterclockwise	$(x, y) \to (-y,\ x)$	$(-3, 2)$	swap, then change the **first** sign
Rotate 180°	$(x, y) \to (-x,\ -y)$	$(-2, -3)$	both signs change
Rotate 270° counterclockwise	$(x, y) \to (y,\ -x)$	$(3, -2)$	swap, then change the **second** sign

Rotations turn about the **origin**. 270° counterclockwise = 90° clockwise. Counterclockwise is the usual direction.

3 Sequences: Prove Two Figures Are Congruent



Goal: map the shaded triangle onto the dashed one.

1. **Reflect** over the y-axis: $(x, y) \to (-x, y)$ (dotted)
2. **Translate** down 4: $(x, y) \to (x, y - 4)$

Each step is rigid, so the triangles are **congruent**.

Two primes (A'') means two moves. Order can matter: do the steps in order!

Dilations & Similarity

NC.8.G.3 • NC.8.G.4

4 Dilations: Same Shape, New Size

Rule (center at the origin):

$$(x, y) \to (kx,\ ky)$$

k = **scale factor**. Multiply **both** coordinates by k.

$k > 1$	**enlargement** (bigger)
$0 < k < 1$	**reduction** (smaller)
$k = 1$	same size

Stays the same: angle measures.
Changes: side lengths (all by k).

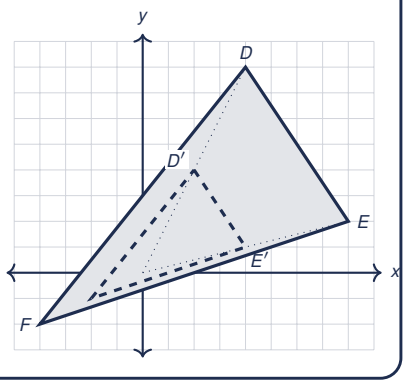
Dilate triangle DEF by $k = \frac{1}{2}$

$D(4, 8) \to D'\left(\frac{1}{2} \cdot 4,\ \frac{1}{2} \cdot 8\right) = (2, 4)$

$E(8, 2) \to E'(4, 1)$

$F(-4, -2) \to F'(-2, -1)$

Each side of $D'E'F'$ is **half** as long.



5 Find the Scale Factor

$$k = \frac{\text{image length}}{\text{original length}}$$

Ex: A side of length 6 becomes 9.

$k = \frac{9}{6} = \frac{9 \div 3}{6 \div 3} = \frac{3}{2}$ → enlargement

Ex: $A(6, 3) \to A'(2, 1)$

$k = \frac{2}{6} = \frac{1}{3}$ → reduction

Find a missing side

Similar triangles. Sides 4 and 6 map to 10 and x.

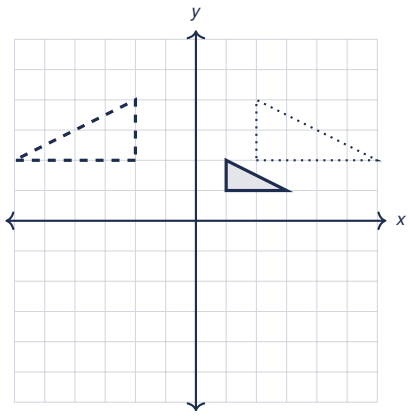
1. $k = \frac{10}{4} = 2.5$
2. $x = 6 \times 2.5$

$\boxed{x = 15}$

Check: $\frac{15}{6} = 2.5$ ✓

$$\begin{array}{r} 2.5 \\ \times \quad 6 \\ \hline 1\ 5.0 \end{array}$$

6 Similar Figures: Prove It with a Sequence



Goal: map the small shaded triangle onto the dashed one.

1. **Dilate** by $k = 2$: $(x, y) \to (2x, 2y)$ (dotted)
2. **Reflect** over the y-axis: $(x, y) \to (-x, y)$

A dilation is in the sequence, so the triangles are **similar** ($\sim$).

Similar: angles are **equal**, sides are **proportional**. (AA works too: two equal angles → similar.)

7 Congruent or Similar?

	Moves used	What stays the same	Symbol
Congruent	translations, reflections, rotations **only**	same size **and** same shape	$\cong$
Similar	any of those **plus a dilation**	same shape, size can change	$\sim$